

GREATER CAIRO REAL ESTATE ANALYSIS

Apartment Pricing Factors & Market Inefficiency Detection

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Introduction

This project analyzes apartment prices in **Greater Cairo** using a dataset of **4,784 properties**. The objective is to understand the key factors affecting price per square meter and identify pricing inefficiencies that could aid data-driven real estate decisions.

Research Question

What factors influence **price per square meter**, and can we detect **overpriced** or **underpriced** apartments in the Greater Cairo market?

Data Preparation

4,784 apartment records collected Missing values handled Outliers removed using IQR method Features: location, view, rooms, floor, area

Location

H₀: No relationship exists between location and price per m².
H₁: Location significantly affects price per m², and is the dominant pricing driver.

Property Features

H₀: No relationship exists between apartment features and price per m².
H₁: Features (view, rooms, floor level) significantly affect price per m².

Market Pricing

H₀: Apartment pricing is random with no detectable pattern or inefficiency.
H₁: Pricing follows predictable patterns and market inefficiencies can be detected.

EXPLORATORY DATA ANALYSIS & RESULTS

PRICE DISTRIBUTION

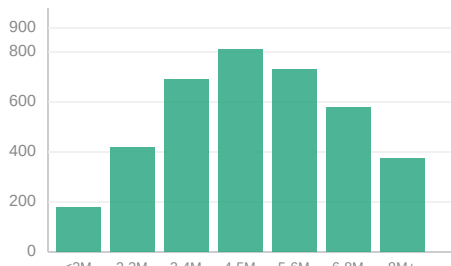


Fig 1 – Median ≈ 4.8M EGP

LOCATION VS PRICE/M²

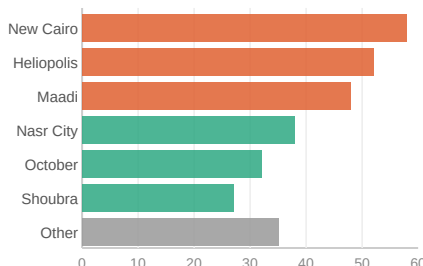


Fig 2 – Location drives pricing

ROOMS VS PRICE/M²

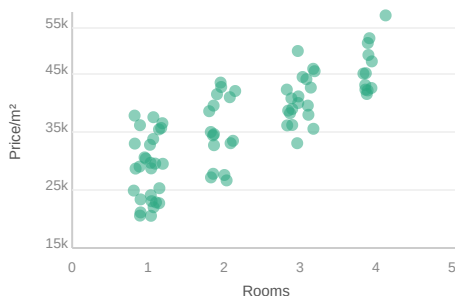


Fig 3 – Room count correlation

VIEW TYPE IMPACT

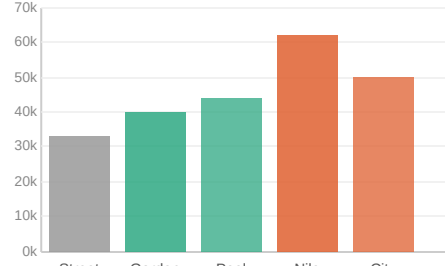


Fig 4 – View premium effect

FLOOR LEVEL VS PRICE/M²

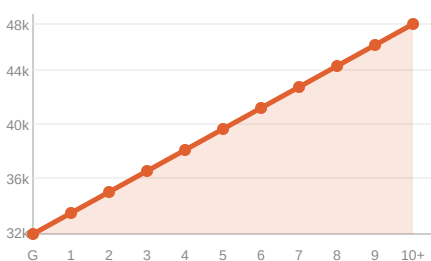


Fig 5 – Higher floors, higher prices

AREA VS TOTAL PRICE

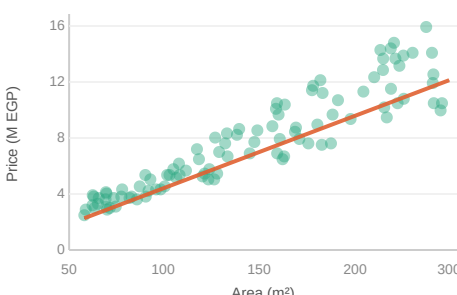


Fig 7 – R² = 0.277 (moderate fit)

MARKET SEGMENTATION

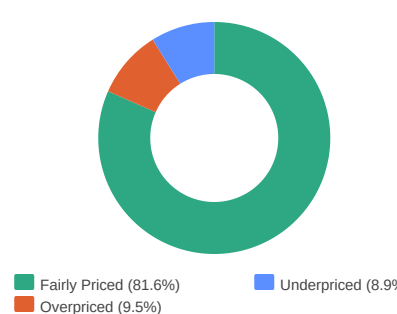


Fig 10 – Pricing fairness distribution

4,784

Apartments Analyzed

4.8M

Median Price (EGP)

37,020

Median Price/m² (EGP)

p < 0.05

All Hypothesis Tests

R²=0.277

Regression Model

81.6%

Fairly Priced Units

ANALYSIS REACHED

1. Location & Key Factors

- Location** is the dominant factor driving price per m² — neighborhoods show markedly different median prices.
- View type** adds a measurable premium; premium views (garden, Nile, city) correlate with higher prices.
- Room count** has a positive association with price per m², though weaker than location.
- All hypothesis tests returned **p-values < 0.05**, rejecting H₀ and confirming that features significantly affect pricing.

2. Regression & Modeling

- A regression model was built to predict price per m² from apartment features.
- R² = 0.277** indicates moderate predictive power — multiple variables jointly influence pricing.
- Floor level shows a positive trend: higher floors generally command higher prices per m².
- Area vs. total price shows a positive linear relationship, confirmed by the regression model (Fig 7).
- Correlation heatmap revealed the strongest feature correlations for location-related variables.

3. Market Inefficiencies

- 81.6%** of apartments are fairly priced relative to comparable units in the dataset.
- 9.5%** are classified as *overpriced* — listed above their predicted market value.
- 8.9%** are classified as *underpriced* — listed below their predicted market value, representing potential buying opportunities.
- Market inefficiencies confirm that pricing is **not random** and is strongly predictable with data-driven tools.

Conclusion

Location is the single most influential factor in Greater Cairo apartment pricing. The analysis proves that pricing is not random and can be predicted with statistical models. With 18.4% of the market showing inefficiency (overpriced or underpriced units), there are **strong opportunities for data-driven investment decisions**. Future work should incorporate more features (building age, amenities, proximity to transport) to improve the R² beyond 0.277 and enhance market segmentation precision.